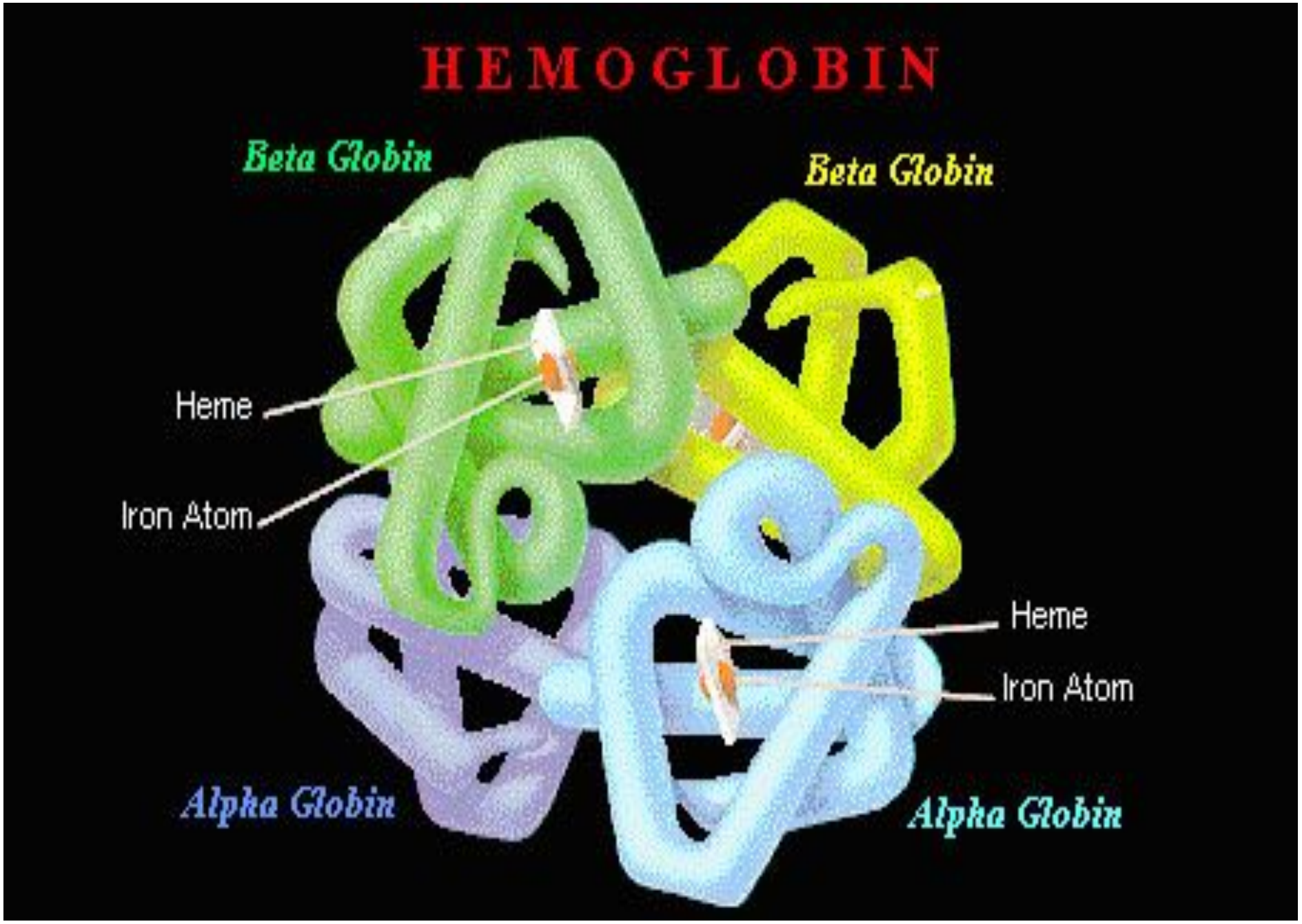
Medical Physics Department
Physiology Lab.
2nd Class
2025-2026
Lab6



Haemoglobin Estimation

(Measurement of Hb concentration by Sahli method)


HEMOGLOBIN
Beta Globin
Beta Globin
Heme
Iron Atom
Heme
Iron Atom
Alpha Globin
Alpha Globin

Hemoglobin

It is the main constituent of RBCs/which gives the blood its characteristic red color.

Normal level:

Women: 12 to 16 g/dl

Men: 13 to 18 g/dl

Children: 11 to 16 g/dl

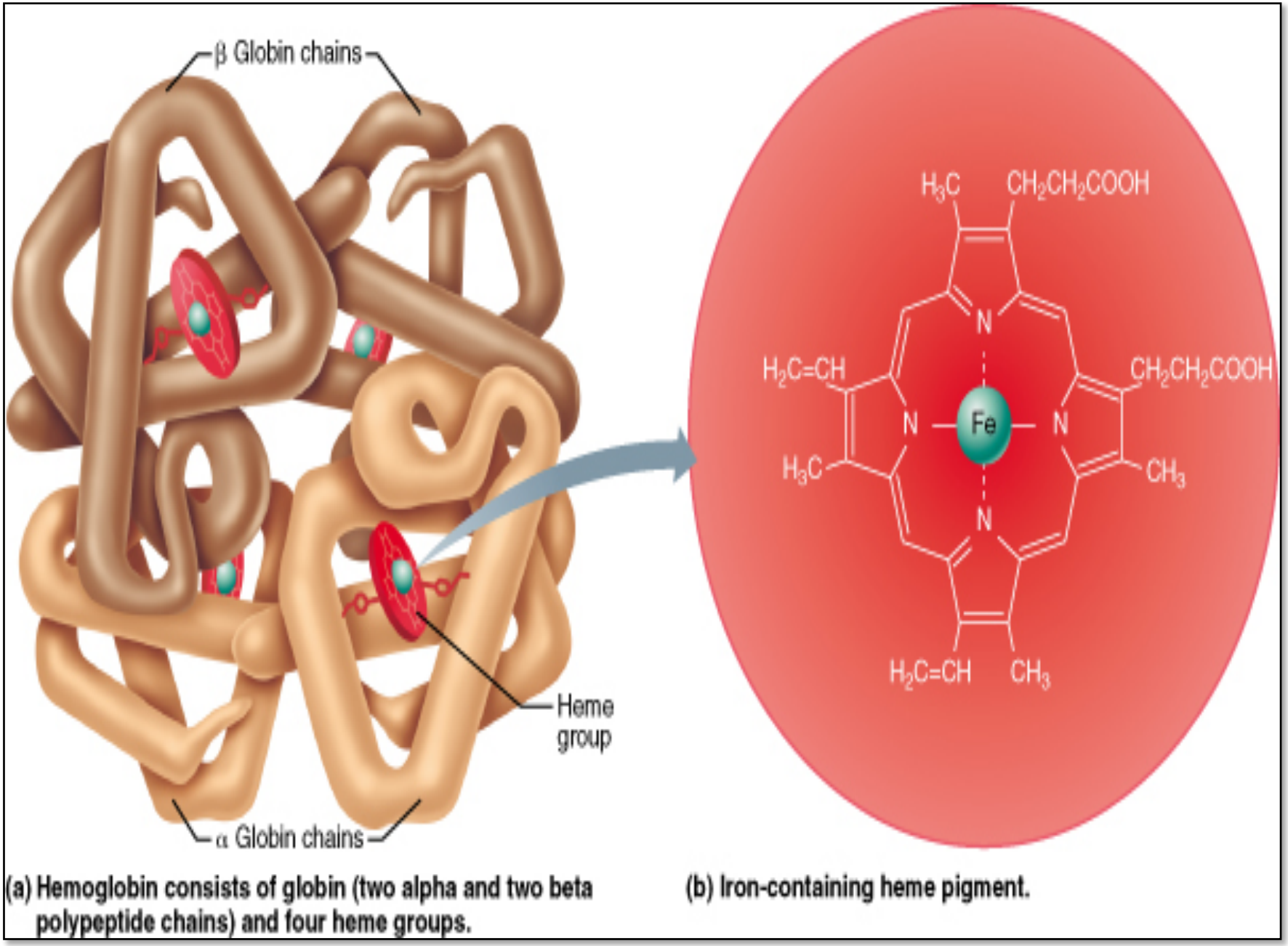
Pregnant women: 11 to 12 g/dl

HB consists of two parts:

1- Pigment heme (iron containing pigment), which constitutes (4%)

2- Protein globin (96%)

Hemoglobin consists of 4 heme groups, each is protein chain (2 pairs of polypeptides in each), 2 of those protein chains form hemoglobin molecule.



(a) Hemoglobin consists of globin (two alpha and two beta polypeptide chains) and four heme groups.

(b) Iron-containing heme pigment.

human hemoglobin (Hb_A)

- In normal adults, human hemoglobin (Hb_A) the 2 types of polypeptides are called α-chains, each of which contains (141 amino acids) and the β-chains, each of which contains (146 amino acids), so the HbA is designated ($\alpha_2\beta_2$).
- Not all the hemoglobin in the normal adult is Hb_A; about 2.5% of hemoglobin is Hb_{A2}, in which the beta chains are replaced by delta (δ) chains ($\alpha_2\delta_2$), where delta contains (146 amino acids) residues, but the last 10 contain individual residues differ from those in the β chain.

Hemoglobin of the fetus

the blood of the human fetus contains fetus Hb (Hb_f), its structure similar to that of HbA except that the beta chains are replaced by gamma (ɣ) chains, so Hb_f is ($\alpha_2\gamma_2$), and fetal Hb is replaced by adult Hb soon after birth.

Function of Hb:

- Act as O2 carriers from the lungs to tissues and back transport of CO2 from tissues to lungs
- Acts as buffer that helps regulate the PH of the blood
- When it is broken down, it forms a bile pigment (which has a role in the digestion of lipids)

Hb can be measured by any of the following methods

a) Colorimetric method – based on color

b) Physical method – based on specific gravity

c) Chemical method – based on iron content of Hb.

d) Gasometric method – based on oxygen combining capacity of Hb.

Significance of Hb estimation

- Decrease in the haemoglobin below the normal range is an indication of anemia.
- Causes for increase in the haemoglobin concentration:
  - Hypoxic states
  - Increased secretion of erythropoetin
  - Polycythemia vera

Sahli´s Kit (Haemometer)



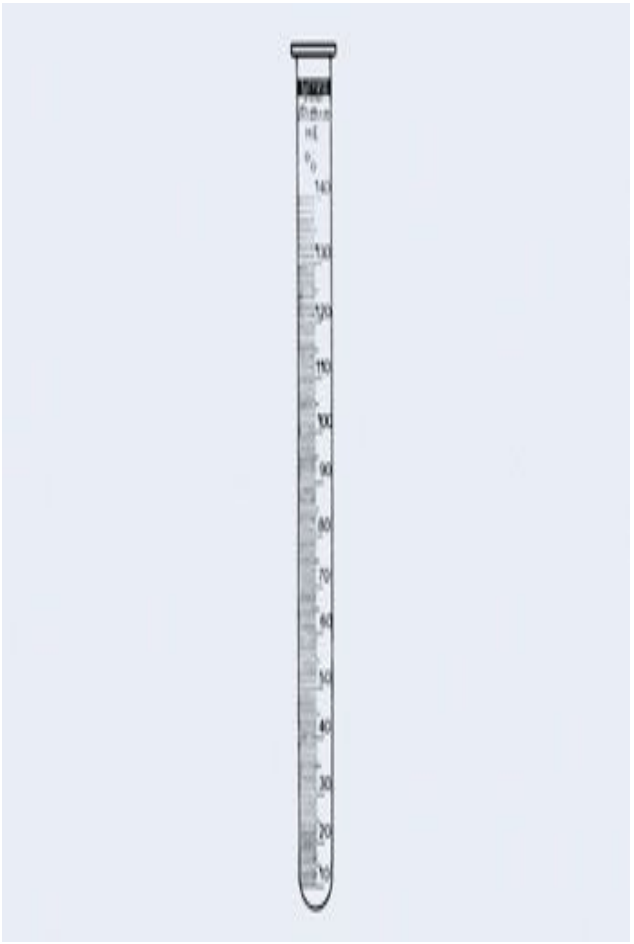
Droppe
r



Sahli's paipette



Hemometer comparative tubes with double scales (g%) & (percent of normal)



Color comparator



Principle Sahli method:

- In this method, the blood Hb is converted to brownish hematin compound by the action of HCl. Each type of Hb must be converted to standard form and color which is and hematin.
- The higher the Hb conc. The intense the color of hematin will be, the intensity of the color is measured comparing it with standard solution of Hb.

Procedure

- Place 5 drops of (0.1N HCl) in the bottom of graduated Sahli tube. This amount should fill the tube to around the (10%) mark on the red scale (red calibration)
- To this add blood sucked till the specific mark (20 µl = 0.02ml) on the Hb pipette. Do not allow air to enter the pipette column
- Insert the tip of the pipette beneath the surface of Sahli tube and gently blow out the blood
- Mix the blood with 0.1N HCl, and by sucking the mixture and expelling it again, and then let the tube stand for 10 min (Hb converts to hematin)
- Add DW to the hematin solution, drop by drop (stir after each addition) until its color matches the color of the standard color


Thank
You